



Lian Cataytay

Data Analyst | Applied Economics Analyst

Davao City, Philippines | 0995 345 3567 | liancataytayjk@gmail.com

Portfolio: liancataytayjk-coder.github.io | GitHub: github.com/liancataytayjk-coder | LinkedIn: linkedin.com/in/liancataytay

SUMMARY

Data analyst and applied economics professional with 3+ years of experience across business analytics, research data workflows, dashboard reporting, statistical modeling, financial analysis, marketing analytics, and risk analysis. Skilled in Excel, SQL, Python, R, Stata, Power BI, Tableau, econometrics, KPI reporting, and translating technical outputs into decision-ready insights for stakeholders.

CORE SKILLS

Analytics: Data Cleaning, EDA, Statistical Analysis, Regression, KPI Reporting, Dashboard Reporting, Data Visualization
Tools: Excel, Google Sheets, SQL, Python, Pandas, Scikit-learn, SHAP, R, RStudio, Stata, Power BI, Tableau
Business/Economics: Financial Analysis, Marketing Analytics, Funnel Analysis, Customer Retention, Risk Modeling, Time-Series Econometrics

EXPERIENCE

Data Analyst Intern | Socio-Economic Research and Data Analytics Center | Aug 2025 - Oct 2025

- Completed a 3-month analytics internship supporting research data cleaning, demographic dataset preparation, and evidence-based reporting workflows.
- Standardized multi-source demographic data using RStudio and Excel, improving analysis readiness, variable consistency, and review efficiency.
- Produced charts, summary tables, and Excel dashboards to communicate key trends for internal reviews and external stakeholder reporting.
- Converted statistical outputs into concise briefs for policymakers, program staff, and community stakeholders, strengthening non-technical decision support.

Freelance Data Analyst | Self-employed | Oct 2024 - Present

- Built monthly revenue and expense trackers with automated budget-vs-actual summaries, variance checks, and leadership-style review tables.
- Designed KPI dashboards covering 8+ metrics, including CTR, conversion rate, ROAS, CAC, ROI, revenue, costs, margins, and funnel movement.
- Unified marketing and sales datasets across the full funnel, from impressions to paid customers, enabling weekly performance monitoring.
- Modeled financial scenarios and A/B test lift using R and regression-style methods to support cost-efficiency and budget-allocation recommendations.

Econometrics and Applied Economics Research Analyst | University of Southeastern Philippines | Aug 2024 - Jan 2026

- Built Prais-Winsten and time-series regression models for macroeconomic series, including GVA, energy use, and technology investment indicators.
- Applied ADF tests, dynamic modeling, and scenario analysis to assess inflation responses to interest-rate shocks and monetary indicators.
- Developed Logit and Probit models for classification-style analysis, validating outputs with confusion matrices, sensitivity, and specificity metrics.
- Created logistic loan-delinquency risk models using borrower, interest-rate, and insurance variables; synthesized policy findings for labor-market research.

Lead Feature Editor | The Collegiate Headlight, USEP | Aug 2022 - Aug 2024

- Wrote, edited, and quality-checked 100+ published pieces across 4 semesters for a university-wide editorial operation.
- Managed content pipelines, deadlines, layout cycles, and revisions, coordinating recurring deliverables across multiple production stages.
- Mentored junior writers on evidence use, narrative structure, source handling, and revision quality to improve publication standards.
- Translated complex institutional and policy issues into accessible narratives, strengthening audience-facing communication and analytical clarity.

Administrative Assistant | Saint John Marie Vianney Academy of Sulop, Inc. | Aug 2020 - Apr 2022

- Managed accounting logs, class records, schedules, correspondence, and internal reports across a 20-month administrative support period.
- Maintained complete operational records during blended learning, supporting continuity across documentation, scheduling, and reporting processes.
- Designed and produced 50+ educational and promotional materials, including worksheets, modules, learning kits, and communication assets.
- Supported school-wide logistics, record keeping, and deadline-driven follow-through for academic, administrative, and stakeholder-facing activities.

PROJECTS

AI Customer Churn Intelligence System

- Built explainable churn prediction using Python, Pandas, Scikit-learn, SHAP, and Power BI to identify high-risk customers.
- Generated churn-risk scores, model-explainability visuals, and business summaries to support targeted retention decision-making.
- Translated predictive outputs into customer-retention insights across modeling, dashboarding, interpretation, and stakeholder reporting.

E-commerce Customer Retention Analytics

- Analyzed customer behavior, transaction patterns, retention signals, and churn risk using Python, Pandas, Tableau, and Excel.
- Built segmentation outputs identifying high-value customers, revenue concentration, repeat-purchase patterns, and retention opportunities.
- Developed Tableau dashboards summarizing customer cohorts, purchase behavior, and revenue patterns for business review workflows.

Banking Risk and Financial Performance Analysis

- Developed credit-risk analytics using Python, Pandas, Excel, and machine learning to evaluate default propensity and portfolio exposure.
- Produced risk-scored loan datasets, feature-importance outputs, and portfolio summaries for loan-performance and credit-risk review.
- Achieved approximately 0.72 ROC AUC in default-risk classification, demonstrating practical model validation and risk interpretation.

Philippine Emigration ARDL Macroeconomic Analysis

- Applied ARDL bounds testing and error-correction modeling in Stata and Excel to assess Philippine emigration trends.
- Modeled short-run and long-run relationships between macroeconomic indicators and emigration using time-series econometrics.
- Structured econometric outputs into a policy-oriented workflow for documentation, interpretation, and decision-focused reporting.

Agricultural SCM Productivity Analysis

- Analyzed agricultural supply-chain management, labor productivity, land productivity, and total factor productivity indicators.
- Built reproducible R Markdown workflows using R and Excel for data preparation, statistical analysis, and documentation.
- Interpreted productivity results for applied agricultural economics, linking supply-chain measures to operational and policy implications.

EDUCATION

University of Southeastern Philippines (USEP) - Davao City

Bachelor of Science in Economics, Development Economics | CGPA: 1.30

Relevant Coursework: Econometrics, Time-Series Analysis, Financial Economics, Development Economics, Statistics, Project Planning and Management, Project Evaluation.